Lecture 21

Type C fundamental representations

Objectives. *A symplectic form on $V = \mathbb{C}^{2n}$ gives an $\mathfrak{sl}_2$ acting on ΛV. Its primitive forms realise the fundamental modules of $\mathfrak{sp}_{2n}$.*

By the end of this lecture you will be able to:

- Compute the C_n roots, coroots and fundamental weights of $\mathfrak{sp}_{2n}$.
- Decompose $\Lambda^k V$ into primitive pieces using the Lefschetz $\mathfrak{sl}_2$.
- Prove $\Lambda^k_0 V \cong L(\omega_k)$ for $1 \le k \le n$.
- Identify $\Lambda^2_0 \mathbb{C}^4$ with the vector module of $\mathfrak{so}_5$.

Throughout, $n \ge 1$ and $V = \mathbb{C}^{2n}$. All modules are finite-dimensional over $\mathbb{C}$.

21.1 The algebra $\mathfrak{sp}_{2n}$ and its C_n data

Write the standard basis of V as $e_1, \dots, e_n, f_1, \dots, f_n$, where f_i is the $(n+i)$-th standard vector. The *symplectic form* is

$$\omega(u, v) = u^{\mathsf{T}} J v, \qquad J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}.$$

Thus

$$\omega(e_i, f_j) = \delta_{ij} = -\omega(f_j, e_i), \qquad \omega(e_i, e_j) = \omega(f_i, f_j) = 0.$$

Definition 21.1 (Symplectic Lie algebra)**.** The *symplectic Lie algebra* is

$$\mathfrak{sp}_{2n} = \{X \in \mathfrak{gl}_{2n}(\mathbb{C}) : \omega(Xu, v) + \omega(u, Xv) = 0 \text{ for all } u, v \in V\}.$$

As in Lecture 10, $\mathfrak{sp}_{2n}$ is a subalgebra. Since $\omega(Xu, v) = u^{\mathsf{T}} X^{\mathsf{T}} J v$, the defining condition is $X^{\mathsf{T}} J + J X = 0$. In $n \times n$ blocks A, B, C, D of X, it says $D = -A^{\mathsf{T}}$, $B = B^{\mathsf{T}}$ and $C = C^{\mathsf{T}}$. Hence

$$\mathfrak{sp}_{2n} = \left\{ \begin{pmatrix} A & B \\ C & -A^{\mathsf{T}} \end{pmatrix} : B = B^{\mathsf{T}},\ C = C^{\mathsf{T}} \right\}, \qquad \dim \mathfrak{sp}_{2n} = 2n^2 + n.$$

Let $\mathfrak{h}$ consist of the diagonal matrices $H(t) = \operatorname{diag}(t_1, \dots, t_n, -t_1, \dots, -t_n)$, and put $\varepsilon_i(H(t)) = t_i$. With matrix units E_{ab}, the bracket $[H(t), E_{ab}]$ is $(d_a - d_b)E_{ab}$, where d is the diagonal of $H(t)$. This gives root vectors:

$$\begin{aligned} X_{ij} &= E_{ij} - E_{n+j,n+i} \quad (i \ne j), && \text{root } \varepsilon_i - \varepsilon_j, \\ Y_{ij} &= E_{i,n+j} + E_{j,n+i} \quad (i < j), && \text{root } \varepsilon_i + \varepsilon_j, \\ Y_{ii} &= E_{i,n+i}, && \text{root } 2\varepsilon_i. \end{aligned}$$

The transposes $Y_{ij}^{\mathsf T}$ and $Y_{ii}^{\mathsf T}$ have roots $-\varepsilon_i-\varepsilon_j$ and $-2\varepsilon_i$. These $2n^2$ matrices and $\mathfrak h$ span $\mathfrak{sp}_{2n}$. So $\mathfrak h$ is its own centraliser, and we have the root decomposition of Lecture 7 with

$$R=\{\pm\varepsilon_i\pm\varepsilon_j : i<j\}\cup\{\pm2\varepsilon_i\}.$$

Take $R_+=\{\varepsilon_i-\varepsilon_j,\ \varepsilon_i+\varepsilon_j\ (i<j),\ 2\varepsilon_i\}$ and $\alpha_i=\varepsilon_i-\varepsilon_{i+1}$ for $i<n$, $\alpha_n=2\varepsilon_n$. These simple roots form the C_n base proved in Lecture 10.

For $\alpha\in R_+$ the coroot $h_\alpha\in[\mathfrak g_\alpha,\mathfrak g_{-\alpha}]$ satisfies $\alpha(h_\alpha)=2$ (Lecture 8). Matrix multiplication gives

$$\begin{aligned}h_{\varepsilon_i-\varepsilon_j}&=[X_{ij},X_{ji}]=E_{ii}-E_{jj}-E_{n+i,n+i}+E_{n+j,n+j},\\ h_{\varepsilon_i+\varepsilon_j}&=[Y_{ij},Y_{ij}^{\mathsf T}]=E_{ii}+E_{jj}-E_{n+i,n+i}-E_{n+j,n+j},\\ h_{2\varepsilon_i}&=[Y_{ii},Y_{ii}^{\mathsf T}]=E_{ii}-E_{n+i,n+i}.\end{aligned}$$

For instance,

$$X_{ij}X_{ji}=E_{ii}+E_{n+j,n+j},\qquad X_{ji}X_{ij}=E_{jj}+E_{n+i,n+i}.$$

For $\mu=\sum_i\mu_i\varepsilon_i$,

$$\mu(h_{\varepsilon_i-\varepsilon_j})=\mu_i-\mu_j,\qquad \mu(h_{\varepsilon_i+\varepsilon_j})=\mu_i+\mu_j,\qquad \mu(h_{2\varepsilon_i})=\mu_i. \tag{21.1}$$

With $h_i=h_{\alpha_i}$, (21.1) gives $\omega_k(h_i)=\delta_{ik}$ for

$$\omega_k=\varepsilon_1+\cdots+\varepsilon_k\quad(1\le k\le n),\qquad \rho=\sum_{k=1}^{n}\omega_k=\sum_{i=1}^{n}(n+1-i)\varepsilon_i.$$

The vectors e_i and f_i have weights ε_i and $-\varepsilon_i$.

Reading off columns, the root vectors satisfy

$$X_{ij}e_j=e_i\ (i<j),\qquad X_{ij}e_l=0\ (l\ne j),\qquad Y_{ij}e_l=0\ (i\le j). \tag{21.2}$$

So every positive root vector maps e_j into $\operatorname{span}\{e_i : i<j\}$.

21.2 The invariant bivector and the contraction

Let $\Lambda V=\bigoplus_{k=0}^{2n}\Lambda^kV$, with $\Lambda^jV=0$ for $j<0$. The quotient construction of Exercise 18.2(i) gives the action on Λ^kV

$$\rho_\wedge(X)(v_1\wedge\cdots\wedge v_k)=\sum_{j=1}^{k}v_1\wedge\cdots\wedge Xv_j\wedge\cdots\wedge v_k.$$

Hence $\rho_\wedge(X)$ satisfies the derivation rule

$$\rho_\wedge(X)(\xi\wedge\eta)=\rho_\wedge(X)\xi\wedge\eta+\xi\wedge\rho_\wedge(X)\eta.$$

On $\Lambda^{2n}V$, only diagonal entries of X survive in the basis wedge. Thus $\rho_\wedge(X)$ acts by their sum $\operatorname{tr}X=0$ for $X\in\mathfrak{sp}_{2n}$.

Definition 21.2 (Bivector and wedge operator)**.** The *symplectic bivector* and the *wedge operator* are

$$\Omega=\sum_{i=1}^{n}e_i\wedge f_i\in\Lambda^2V,\qquad L\colon\Lambda V\to\Lambda V,\quad L(\eta)=\Omega\wedge\eta.$$

Lemma 21.3. *For $X \in \mathfrak{sp}_{2n}$, $\rho_\wedge(X)\Omega = 0$. Hence L commutes with every $\rho_\wedge(X)$.*

Proof. The columns of X are

$$Xe_j = \sum_i (A_{ij}e_i + C_{ij}f_i), \qquad Xf_j = \sum_i (B_{ij}e_i - A_{ji}f_i).$$

Therefore

$$\begin{aligned}\rho_\wedge(X)\Omega &= \sum_j (Xe_j \wedge f_j + e_j \wedge Xf_j) \\ &= \sum_{i,j} \big(A_{ij}\, e_i \wedge f_j - A_{ji}\, e_j \wedge f_i\big) + \sum_{i,j} C_{ij}\, f_i \wedge f_j + \sum_{i,j} B_{ij}\, e_j \wedge e_i.\end{aligned}$$

The A-terms cancel after exchanging i and j in the second term. The C- and B-sums vanish, since symmetric coefficients meet antisymmetric wedges.

The derivation rule now gives $\rho_\wedge(X)(\Omega \wedge \eta) = \Omega \wedge \rho_\wedge(X)\eta$. ∎

For $\varphi \in V^*$, the *interior product* is

$$\iota_\varphi(v_1 \wedge \cdots \wedge v_k) = \sum_{j=1}^{k} (-1)^{j-1} \varphi(v_j)\, v_1 \wedge \cdots \wedge \widehat{v_j} \wedge \cdots \wedge v_k.$$

The right side is alternating and multilinear, so ι_φ is well defined. For $v \in V$, let $m_v \colon \Lambda V \to \Lambda V$ be given by $m_v(\eta) = v \wedge \eta$. Then

$$m_u m_v + m_v m_u = 0, \qquad \iota_\varphi \iota_\psi + \iota_\psi \iota_\varphi = 0, \qquad \iota_\varphi m_v + m_v \iota_\varphi = \varphi(v)\,\mathrm{id}_{\Lambda V}. \tag{21.3}$$

The third identity is the defining formula with $v_1 = v$. For the second, in $\iota_\varphi \iota_\varphi$ the terms removing v_p then v_q, and v_q then v_p, have opposite signs. So $\iota_\varphi^2 = 0$, and expanding $\iota_{\varphi+\psi}^2 = 0$ gives anticommutation.

Let (e_i^*, f_i^*) be the dual basis. As a bilinear form,

$$\omega(u, v) = \sum_i \big(e_i^*(u) f_i^*(v) - f_i^*(u) e_i^*(v)\big).$$

Definition 21.4 (Contraction and primitive forms)**.** The *contraction* is the map

$$\iota_\omega = \sum_{i=1}^{n} \iota_{f_i^*} \iota_{e_i^*} \colon \Lambda^k V \to \Lambda^{k-2} V.$$

The space of *primitive k-forms* is

$$\Lambda_0^k V = \ker\big(\iota_\omega \colon \Lambda^k V \to \Lambda^{k-2} V\big).$$

For $u, v \in V$,

$$\begin{aligned}\iota_\omega(u \wedge v) &= \sum_i \iota_{f_i^*}\big(e_i^*(u)v - e_i^*(v)u\big) \\ &= \sum_i \big(e_i^*(u) f_i^*(v) - e_i^*(v) f_i^*(u)\big) = \omega(u, v).\end{aligned}$$

Lemma 21.5. *The contraction ι_ω commutes with every $\rho_\wedge(X)$, $X \in \mathfrak{sp}_{2n}$. Hence each $\Lambda_0^k V$ is a submodule.*

Proof. First, for $\varphi \in V^*$,

$$\iota_\varphi \, \rho_\wedge(X) = \rho_\wedge(X) \, \iota_\varphi + \iota_{\varphi \circ X}. \tag{21.4}$$

Apply the left side to $v_1 \wedge \cdots \wedge v_k$. This gives the sum over l of $\iota_\varphi(v_1 \wedge \cdots \wedge Xv_l \wedge \cdots \wedge v_k)$. Removing position l gives $(-1)^{l-1}\varphi(Xv_l)$ times the monomial without v_l. These terms sum to $\iota_{\varphi \circ X}$. Removing a position $j \neq l$ gives $(-1)^{j-1}\varphi(v_j)$ times the monomial without v_j and with Xv_l in place of v_l. Summed over l and $j \neq l$, these are the terms of $\rho_\wedge(X)\iota_\varphi(v_1 \wedge \cdots \wedge v_k)$.

Second, rows i and $n+i$ of X give

$$e_i^* \circ X = \sum_j (A_{ij} e_j^* + B_{ij} f_j^*), \qquad f_i^* \circ X = \sum_j (C_{ij} e_j^* - A_{ji} f_j^*).$$

Using (21.4) twice,

$$\begin{aligned} [\iota_\omega, \rho_\wedge(X)] &= \sum_i \left(\iota_{f_i^* \circ X} \iota_{e_i^*} + \iota_{f_i^*} \iota_{e_i^* \circ X} \right) \\ &= \sum_{i,j} \left(C_{ij} \iota_{e_j^*} \iota_{e_i^*} - A_{ji} \iota_{f_j^*} \iota_{e_i^*} + A_{ij} \iota_{f_i^*} \iota_{e_j^*} + B_{ij} \iota_{f_i^*} \iota_{f_j^*} \right). \end{aligned}$$

The A-terms cancel after exchanging i and j. The C- and B-terms vanish by symmetry and (21.3). ∎

21.3 The Lefschetz $\mathfrak{sl}_2$ and primitive decomposition

Let H act on $\Lambda^k V$ as multiplication by $n - k$.

Proposition 21.6 (Lefschetz relation). *On ΛV,*

$$[\iota_\omega, L] = H, \qquad [H, \iota_\omega] = 2\iota_\omega, \qquad [H, L] = -2L.$$

Hence $e \mapsto \iota_\omega$, $h \mapsto H$, $f \mapsto L$ is a representation of $\mathfrak{sl}_2$ on ΛV. These three operators commute with every $\rho_\wedge(X)$, $X \in \mathfrak{sp}_{2n}$.

Proof. For $\eta \in \Lambda^k V$,

$$[H, L]\eta = (n - k - 2)L\eta - (n - k)L\eta = -2L\eta.$$

The same calculation, with degree $k - 2$, gives $[H, \iota_\omega] = 2\iota_\omega$.

For the first relation, put $a_i = m_{e_i}$, $b_i = m_{f_i}$, $a_i^* = \iota_{e_i^*}$ and $b_i^* = \iota_{f_i^*}$. Then

$$L = \sum_i a_i b_i, \qquad \iota_\omega = \sum_i b_i^* a_i^*.$$

By (21.3), any two of these operators anticommute, except

$$a_i^* a_i + a_i a_i^* = 1, \qquad b_i^* b_i + b_i b_i^* = 1.$$

If p and q each anticommute with r and s, then pq commutes with rs. Hence $b_j^* a_j^*$ commutes with $a_i b_i$ for $j \neq i$, and

$$[\iota_\omega, L] = \sum_{i=1}^{n} [b_i^* a_i^*, a_i b_i].$$

Fix i and drop the index. Put $N_a = aa^*$ and $N_b = bb^*$. Then N_a commutes with b, b^*, and N_b commutes with a, a^*. So

$$\begin{aligned} b^* a^* a b &= b^*(1 - N_a)b = (1 - N_a)b^* b = (1 - N_a)(1 - N_b), \\ ab\, b^* a^* &= a N_b a^* = N_b\, aa^* = N_a N_b. \end{aligned}$$

Subtracting, $[b_i^* a_i^*, a_i b_i] = 1 - N_{a_i} - N_{b_i}$.

On a wedge of distinct basis vectors, N_{a_i} is 1 if e_i is a factor and 0 otherwise, because $\iota_{e_i^*}$ removes e_i with a sign and m_{e_i} restores it with the same sign. The same holds for N_{b_i} and f_i. Thus $\sum_i (N_{a_i} + N_{b_i}) = k$ on $\Lambda^k V$, and

$$[\iota_\omega, L] = \sum_{i=1}^{n} (1 - N_{a_i} - N_{b_i}) = n - k = H \quad \text{on } \Lambda^k V.$$

Commutation with $\rho_\wedge(X)$ follows from Lemmas 21.3 and 21.5, since H is scalar on each submodule $\Lambda^k V$. ∎

By Lecture 6, a finite-dimensional $\mathfrak{sl}_2$-module is a direct sum of irreducibles V_m. Each V_m has a basis $v_j = f^j v_0$, $0 \le j \le m$, with $hv_j = (m-2j)v_j$ and $fv_m = 0$. Moreover $\ker e \cap V_m = \mathbb{C} v_0$, of h-weight $m \ge 0$.

Theorem 21.7 (Primitive decomposition)**.**

(a) $\Lambda_0^k V = 0$ *for* $k > n$.

(b) If $0 \le r \le n$ *and* $0 \ne \eta \in \Lambda_0^r V$*, then* $L^j \eta \ne 0$ *for* $0 \le j \le n - r$*, and* $L^{n-r+1}\eta = 0$.

(c) For $0 \le k \le 2n$*,*

$$\Lambda^k V = \bigoplus_{j=\max(0,k-n)}^{\lfloor k/2 \rfloor} L^j \big(\Lambda_0^{k-2j} V\big),$$

and each $L^j \colon \Lambda_0^{k-2j} V \to L^j \Lambda_0^{k-2j} V$ *is an isomorphism of* $\mathfrak{sp}_{2n}$*-modules.*

(d) For $0 \le k \le n$*, the map* $\iota_\omega \colon \Lambda^k V \to \Lambda^{k-2} V$ *is onto, and*

$$\dim \Lambda_0^k V = \binom{2n}{k} - \binom{2n}{k-2},$$

with $\binom{2n}{-1} = \binom{2n}{-2} = 0$.

Proof. Decompose $\Lambda V = \bigoplus_s U_s$ into irreducible $\mathfrak{sl}_2$-submodules. Let $U_s \cong V_{m_s}$ have basis $v_{s,j} = L^j v_{s,0}$. The eigenvalues $n - k$ of H are distinct, so the H-eigenspaces are exactly the $\Lambda^k V$. Hence $v_{s,0} \in \Lambda^{r_s} V$ with $m_s = n - r_s$, and $v_{s,j} \in \Lambda^{r_s + 2j} V$.

Since ι_ω preserves each U_s, its kernel is $\bigoplus_s \mathbb{C} v_{s,0}$. Taking degree r, the vectors $v_{s,0}$ with $r_s = r$ form a basis of $\Lambda_0^r V$. As $m_s \ge 0$, every $r_s \le n$. This proves (a).

For (b), write $\eta = \sum c_s v_{s,0}$ over $r_s = r$, so all $m_s = n - r$. Then $L^j \eta = \sum c_s v_{s,j}$. It is nonzero for $j \le n - r$ by independence, and zero for $j = n - r + 1$.

For (c), the homogeneous basis vectors $v_{s,j}$ with $r_s + 2j = k$ form a basis of $\Lambda^k V$. For fixed j they are the images under L^j of the basis of $\Lambda_0^{k-2j} V$, and they exist exactly when $j \le n - (k - 2j)$, that is, $j \ge k - n$. By (b), L^j is injective on $\Lambda_0^{k-2j} V$ in this range. It commutes with $\mathfrak{sp}_{2n}$ by Proposition 21.6.

For (d), compare (c) in degrees k and $k - 2$ when $2 \le k \le n$:

$$\dim \Lambda^k V = \dim \Lambda_0^k V + \sum_{j \ge 1} \dim \Lambda_0^{k-2j} V = \dim \Lambda_0^k V + \dim \Lambda^{k-2} V.$$

Rank and nullity show that ι_ω is onto. For $k = 0, 1$, its target is zero. ∎

21.4 Primitive exterior powers are fundamental

Theorem 21.8. *For $1 \le k \le n$,*

$$\Lambda^k_0 V \cong L(\omega_k), \qquad \dim L(\omega_k) = \binom{2n}{k} - \binom{2n}{k-2} = \frac{2(n+1-k)\,(2n+1)!}{k!\,(2n+2-k)!}.$$

Proof. Step 1: a highest vector. Since $k \le n$, the vector $v_k = e_1 \wedge \cdots \wedge e_k$ is defined. Each $\iota_{e_i^*} v_k$ is a wedge of e's, so $\iota_{f_i^*}$ kills it. Thus $v_k \in \Lambda^k_0 V$.

$$\rho_\wedge(H(t))v_k = (t_1 + \cdots + t_k)v_k,$$

so v_k has weight ω_k. Let X be a positive root vector. By (21.2), Xe_j lies in the span of e_i with $i < j \le k$. Each such e_i is already a factor, so every term of $\rho_\wedge(X)v_k$ vanishes.

Step 2: the generated submodule. Let W be the submodule generated by v_k. By PBW (Lecture 11), $W = U(\mathfrak{n}_-)v_k$. Monomials in negative root vectors lower weights, so $W_{\omega_k} = \mathbb{C}v_k$. The cyclic-module lemma of Lecture 13 applies to this finite-dimensional weight module. Hence $W \cong L(\omega_k)$.

Step 3: the Weyl dimension. By Lecture 17 and (21.1),

$$\dim L(\lambda) = \frac{F(\lambda+\rho)}{F(\rho)}, \qquad F(\mu) = \prod_{\alpha \in R_+} \mu(h_\alpha) = \prod_{i<j} (\mu_i^2 - \mu_j^2) \prod_i \mu_i.$$

For a finite set S of positive integers put $F_S = \prod_{a\in S} a \prod_{a>b\in S}(a^2-b^2)$. The coordinates of ρ form $\{1,\dots,n\}$. Those of $\omega_k + \rho$ are $n+2-i$ for $i \le k$ and $n+1-i$ for $i > k$. They form $S = T \setminus \{m\}$, where $T = \{1,\dots,n+1\}$ and $m = n+1-k$. Both coordinate lists are strictly decreasing, so $F(\rho) = F_{\{1,\dots,n\}}$ and $F(\omega_k+\rho) = F_S$. Then

$$F_T = F_{\{1,\dots,n\}} \cdot (n+1) \prod_{b=1}^{n} \big((n+1)^2 - b^2\big),$$

$$F_T = F_S \cdot m \prod_{b\in T,\, b\neq m} |m-b|\,(m+b).$$

The first product is

$$(n+1)\cdot n! \cdot \frac{(2n+1)!}{(n+1)!} = (2n+1)!.$$

In the second,

$$\prod_{b\neq m} |m-b| = (m-1)!\,(n+1-m)!, \qquad \prod_{b\neq m}(m+b) = \frac{(n+1+m)!}{m!\cdot 2m}.$$

Equating the two expressions for F_T,

$$\dim L(\omega_k) = \frac{F_S}{F_{\{1,\dots,n\}}} = \frac{2m\,(2n+1)!}{(n+1-m)!\,(n+1+m)!} = \frac{2(n+1-k)\,(2n+1)!}{k!\,(2n+2-k)!}.$$

Step 4: comparison. Over the common denominator $k!(2n+2-k)!$,

$$\binom{2n}{k} - \binom{2n}{k-2} = \frac{(2n)!\,\big[(2n+2-k)(2n+1-k) - k(k-1)\big]}{k!\,(2n+2-k)!}.$$

With $N = 2n+1$, the bracket is

$$(N+1-k)(N-k) - k(k-1) = N(N+1-2k) = 2(2n+1)(n+1-k).$$

For $k = 1$, the convention $\binom{2n}{-1} = 0$ agrees with $k(k-1) = 0$. So $\dim W = \dim \Lambda^k_0 V$ by Theorem 21.7(d). Since $W \subseteq \Lambda^k_0 V$, the two are equal. ∎

Theorems 21.7(c) and 21.8, with $\Lambda^0_0 V = \mathbb{C}$, give for $0 \le k \le n$

$$\Lambda^k V \cong \bigoplus_{j=0}^{\lfloor k/2 \rfloor} L(\omega_{k-2j}), \qquad \omega_0 = 0.$$

21.5 The case of $\mathfrak{sp}_4$

Example 21.9 ($\Lambda^2\mathbb{C}^4 = \mathbb{C} \oplus L(\omega_2)$)**.** Let $n = 2$. Order the basis of $\Lambda^2 V$ as

$$e_1 \wedge e_2,\ e_1 \wedge f_1,\ e_1 \wedge f_2,\ e_2 \wedge f_1,\ e_2 \wedge f_2,\ f_1 \wedge f_2.$$

Since $\iota_\omega(u \wedge v) = \omega(u, v)$, the matrix of $\iota_\omega \colon \Lambda^2 V \to \mathbb{C}$, using the basis 1 of $\mathbb{C}$, is

$$D(\iota_\omega) = \begin{pmatrix} 0 & 1 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

Its kernel $\Lambda^2_0 V$ has basis and weights

vector	$e_1 \wedge e_2$	$e_1 \wedge f_2$	$e_1 \wedge f_1 - e_2 \wedge f_2$	$e_2 \wedge f_1$	$f_1 \wedge f_2$
weight	$\varepsilon_1 + \varepsilon_2$	$\varepsilon_1 - \varepsilon_2$	0	$-\varepsilon_1 + \varepsilon_2$	$-\varepsilon_1 - \varepsilon_2$

Also $\iota_\omega \Omega = 2 \neq 0$. Hence

$$\Lambda^2 V = \mathbb{C}\Omega \oplus \Lambda^2_0 V \cong \mathbb{C} \oplus L(\omega_2), \qquad 6 = 1 + 5.$$

Now compare with $\mathfrak{so}_5$. Let $\mathrm{vol} = e_1 \wedge f_1 \wedge e_2 \wedge f_2$. Define the bilinear form $\beta \colon \Lambda^2 V \times \Lambda^2 V \to \mathbb{C}$ by

$$x \wedge y = \beta(x, y)\,\mathrm{vol}.$$

It is symmetric, because 2-forms commute. Write Xx for $\rho_\wedge(X)x$. Since $\rho_\wedge(X) = 0$ on $\Lambda^4 V$, the form β is invariant:

$$0 = \rho_\wedge(X)(x \wedge y) = Xx \wedge y + x \wedge Xy = \big(\beta(Xx, y) + \beta(x, Xy)\big)\mathrm{vol}.$$

Weights add under $\wedge$ and vol has weight 0. So only pairs of opposite weights can pair nontrivially. Reordering wedges gives

$$\begin{aligned} e_1 \wedge e_2 \wedge f_1 \wedge f_2 &= -\mathrm{vol}, \\ e_1 \wedge f_2 \wedge e_2 \wedge f_1 &= -\mathrm{vol}, \\ (e_1 \wedge f_1 - e_2 \wedge f_2) \wedge (e_1 \wedge f_1 - e_2 \wedge f_2) &= -2\,\mathrm{vol}. \end{aligned}$$

Hence β is nondegenerate on $\Lambda^2_0 V$.

The action therefore gives a homomorphism $\mathfrak{sp}_4 \to \mathfrak{so}(\Lambda^2_0 V, \beta)$. Its kernel K is an ideal, stable under $\operatorname{ad}\mathfrak{h}$. So K is the sum of $K \cap \mathfrak{h}$ and the root spaces it contains. The weights $\varepsilon_1 \pm \varepsilon_2$ separate $\mathfrak{h}$, so $K \cap \mathfrak{h} = 0$. If $\mathfrak{g}_\alpha \subseteq K$, then $h_\alpha \in [\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] \subseteq K$, which is impossible. Thus $K = 0$. All nondegenerate symmetric forms on $\mathbb{C}^5$ are equivalent, so $\mathfrak{so}(\Lambda^2_0 V, \beta) \cong \mathfrak{so}_5$. Moreover,

$$\dim \mathfrak{sp}_4 = 10 = \dim \mathfrak{so}_5.$$

Hence $\mathfrak{sp}_4 \cong \mathfrak{so}_5$, and $\Lambda^2_0 \mathbb{C}^4$ becomes the vector module $\mathbb{C}^5$.

In the B_2 coordinates of Lecture 10, put

$$\varepsilon'_1 = \varepsilon_1 + \varepsilon_2, \qquad \varepsilon'_2 = \varepsilon_1 - \varepsilon_2.$$

The weights of $\Lambda^2_0 V$ are then $0, \pm\varepsilon'_1, \pm\varepsilon'_2$, as for the vector module. The B_2 simple roots are

$$\begin{aligned} \varepsilon'_1 - \varepsilon'_2 &= 2\varepsilon_2 = \alpha_2, \\ \varepsilon'_2 &= \varepsilon_1 - \varepsilon_2 = \alpha_1. \end{aligned}$$

Thus the labels are exchanged. The module $L(\omega_2)$ for C_2 is $L(\omega'_1)$ for B_2.

Self-checks

(1) Compute $\dim \Lambda_0^3 \mathbb{C}^8$ for $\mathfrak{sp}_8$.
(2) Why is $\Lambda_0^{n+1} V = 0$, without any dimension count?
(3) For $\mathfrak{sp}_4$, which pieces of Theorem 21.7(c) occur in $\Lambda^4 V$?

Answers. (1) $\binom{8}{3} - \binom{8}{1} = 56 - 8 = 48$. (2) Such a vector is killed by $e = \iota_\omega$ and has h-weight -1. A nonzero highest vector of an $\mathfrak{sl}_2$-module has weight ≥ 0 (Lecture 6). (3) Here $k = 4$, so $j \geq k - n = 2$. Only $L^2 \Lambda_0^0 V$ occurs: $\Lambda^4 V = \mathbb{C}\,\Omega \wedge \Omega$.

Summary

The operators L, ι_ω and H span an $\mathfrak{sl}_2$ commuting with $\mathfrak{sp}_{2n}$. Primitive forms vanish above degree n, and $\Lambda_0^k V \cong L(\omega_k)$ for $1 \leq k \leq n$.

Reading

William Fulton and Joe Harris, *Representation Theory: A First Course*, Lectures 16–17. Roger Carter, *Lie Algebras of Finite and Affine Type*, Sections 13.6–13.7. Exercise Sheet 21 gives further calculations.